

Public Economics (ECON 131)

Section #10: Social Insurance

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Contents

	Page
1 Key Concepts	1
1.1 Insurance Intro	1
1.2 Expected Utility Model	2
2 Examples	3
2.1 Adverse Selection	3
2.2 Moral Hazard	3
3 Practice: Disability Insurance	4

1 Key Concepts

1.1 Insurance Intro

- **Social insurance programs** are government interventions in the provision of insurance against adverse events.
 - Eg: health insurance, disability insurance, unemployment insurance
 - Government provision can address market failures. May also be motivated by desire to redistribute, provide services that are considered a basic right, address individual failures, or reduce administrative costs
- These insurance systems allow for consumption smoothing across states of the world.
- **Insurance premiums:** Money that is paid to an insurer so that an individual will be insured against adverse events.

1.2 Expected Utility Model

- Individuals maximize expected utility (EU) over possible states of the world, where $q = \text{Pr}(\text{adverse event})$:

$$EU = (1-q) \cdot U(\text{consumption with no adverse event}) + q \cdot U(\text{consumption with adverse event})$$

- An insurance premium is **actuarially fair** if it is set equal to the insurer's expected payout.

$$p = qb$$

Components of consumption: Income W (regardless of health), medical cost d if sick, premium p , and benefit b if sick. Therefore, expected utility is given by:

$$EU = (1 - q) \cdot U(W - p) + q \cdot U(W - p - d + b)$$

Steps: Income W , when sick incur medical cost d , premium p , payout b only if sick: $EU = (1 - q)U(W - p) + qU(W - p - d + b)$. Expected profits to insurers = $EP = p - q \cdot b$. Set expected profits to 0 if competition among insurers. $\rightarrow b = p/q$. Intuition for $b = p/q$: insurers get p from every individual, only have to pay out for q of them, so can give p/q benefits to those who they pay out to to break even. Then individual chooses p to maximize: $EU = (1-q)U(W-p) + qU(W-d-p+p/q)$. FOC gives $p = d \cdot q$. Note perfect insurance means their consumption is exactly the same in two states ($W - d \cdot q$)

- Asymmetric information: individuals have private information about their risk factors.
 - **Adverse selection**: individuals with higher risk are more likely to purchase insurance. This makes insurance costlier, and may make the market unravel
 - **Moral hazard**: individuals may change their behavior in response to being insured

2 Examples

2.1 Adverse Selection

Do health insurance and car insurance markets both suffer from adverse selection? Are there reasons why the government should intervene in the health insurance market and not in the car market?

Solution:

- Yes, like most insurance markets, health and car suffer from adverse selection. Poor car drivers and sick people would want insurance the most.
- We have seen in lecture that in addition to dealing with adverse selection the government might want to intervene to
 - (1) Redistribute based on negative health condition as it is seen out of the control of a person and fair
 - (2) To internalize externalities (think vaccines)
 - (3) To prevent individual failures which are more likely in a more complex market like health
 - (4) To lower administrative costs that are huge in private health markets

2.2 Moral Hazard

Unemployment insurance in the US typically pays the unemployed 50% of their previous wage for about 6 months. During the recession of 2009 it was extended temporarily to a year. Why is unemployment insurance far from 100% replacement? Why did government decided to extend UI during the recession?

Solution:

- 100% UI benefits would provide no incentives for job search until the end of their benefit period. This is a form of moral hazard, since in the absence of UI individuals would look fast for a new job and therefore get off the benefits.
- During the recession there were so few jobs available that not finding a job was hardly a sign of moral hazard.

3 Practice: Disability Insurance

Consider an economy where there are three types of people who want to purchase disability insurance. Each type has the same health-based risks, such that they each have a 2 percent chance of being incapacitated due to health risks, which are uncorrelated with their risk taking behavior. However, each type of person differs in their hobbies and work in the following manner:

- Low-risk types walk to work and have very low-risk hobbies. Their outside risk of being incapacitated is 3 percent.
- Medium-risk types drive to work and actively play soccer on the weekends. Therefore, their non-health risk of being incapacitated is 8 percent.
- High-risk types bike to work, work as firefighters, and skydive on weekends. Their outside risk of being incapacitated is 58 percent.

Assume that there are equal numbers of each type. Long-term care insurance provides income if they are incapacitated for the rest of their life (there are no additional costs). Suppose individuals have the following utility function over consumption (or income):

$$u(c) = \log(c)$$

Individuals earn \$500 if healthy, but only \$10 if incapacitated. They make the decision to purchase insurance before the event occurs.

- (a) Show that with competitive insurance markets (and actuarially fair premiums) that any given individual will choose to fully insure. **Solution:**

First off, we know that the individual wants to purchase a policy with premium (or price) p and payout b to maximize their expected utility. Let q be the probability of an individual being incapacitated due to health risks, w the income of the individual when healthy and $w - d$ the income of the individual when incapacitated. Then, the expected utility of the individual (that the individual will maximize) can be written as:

$$E[U] = (1 - q) \log(w - p) + q \log(w - d - p + b)$$

We know that $w = \$500$ and the loss of income due to disability is $d = \$490$, thus $w - d = \$10$. In addition, if insurers are perfectly competitive, then firm profits equal $p - q*b = 0 \Rightarrow p = q*b$. The individual's maximization problem is then:

$$\max_p E[U] = (1 - q) \log(500 - qb) + q \log(10 - qb + b)$$

FOC:

$$\begin{aligned}\frac{\partial E[U]}{\partial b} &= 0 = \frac{-q(1-q)}{(500-qb)} + \frac{q(-1+q)}{10-qb+b} \\ \frac{q(1-q)}{500-qb} &= \frac{q(1-q)}{10-qb+b} \\ b &= 490 = d\end{aligned}$$

- (b) What is the actuarially fair price for the insurance for each group? What is the expected utility of each group given that price if they buy insurance? What is the expected utility if they do not buy insurance? What prices are each groups willing to pay for the insurance? Assume insurers are perfectly competitive.

The actuarially fair optimal premium (and implicit benefit level) is given by: $p = d * \text{Pr}(\text{disability}) = 490 * q$ and $b = d = 490$.

Therefore, the actuarially fair price for insurance for

- the low-risk types is $p = 490 * (0.05) = \$24.5$
- the medium-risk types is: $p = 490 * (0.10) = \$49$
- the high-risk types is: $p = 490 * (0.60) = \$294$

Expected utility of each group if they buy (full) insurance at price p is:

$$E[U] = (1-q) \log(500-p) + q \log(10-p+b)$$

Therefore, the expected utility with insurance for

- the low-risk types is $E(U) = \log(500 - 24.5) = 6.16$
- the medium-risk types is: $E(U) = \log(500 - 49) = 6.11$
- the high-risk types is: $E(U) = \log(500 - 294) = 5.33$

Expected utility of each group if they do not buy insurance is:

$$E[U] = (1-q) \log(500) + q \log(10)$$

Therefore, the expected utility without insurance for

- the low-risk types is $E[U] = (0.95) \log(500) + (0.05) \log(10) = 6.02$
- the medium-risk types is: $E[U] = (0.90) \log(500) + (0.10) \log(10) = 5.82$
- the high-risk types is: $E(U) = (0.40) \log(500) + (0.60) \log(10) = 3.87$

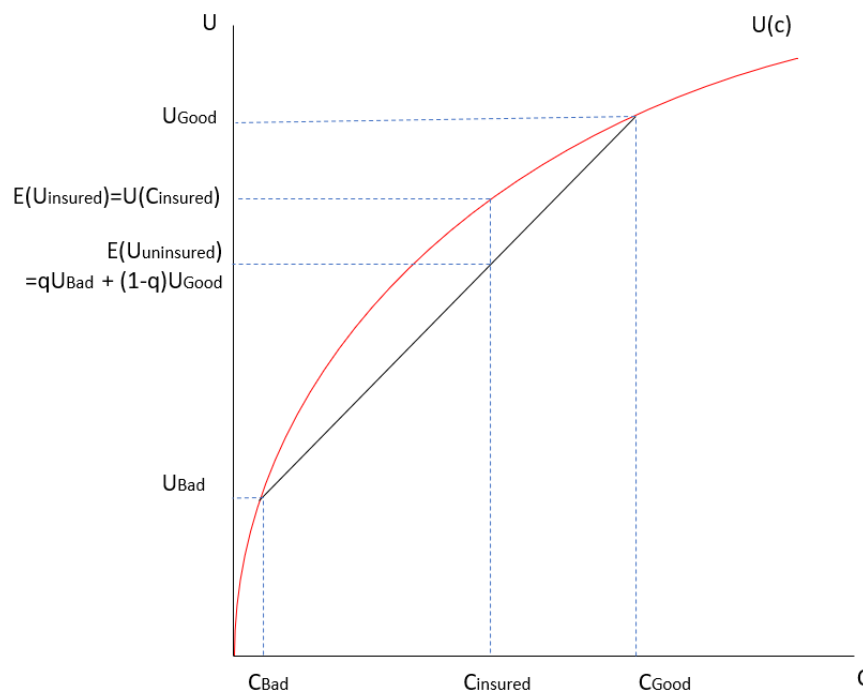
Prices each group is willing to pay for insurance: Finally, to be willing to pay for insurance, the expected utility from life with the insurance must be better than the expected utility from life without the insurance. To simplify, we will assume that individuals purchase enough insurance to fully insure themselves: $U_{ins} = \log(500 - p)$.

$$\begin{aligned}\log(500 - p) &\geq (1 - q) \log(500) + q \log(10) \\ \log(500 - p) &\geq \log(500^{1-q} * 10^q) \\ 500 - p &\geq 500^{1-q} * 10^q \\ p &\leq 500 - 500^{1-q} * 10^q\end{aligned}$$

Therefore, the maximum price for

- the low-risk types is $p \leq 500 - 500^{0.95} * 10^{0.05} = \88.83
- the medium-risk types is: $p \leq 500 - 500^{0.90} * 10^{0.10} = \161.88
- the high-risk types is: $p \leq 500 - 500^{0.40} * 10^{0.60} = \452.18

(c) Show graphically how someone with a log utility (exhibiting diminishing marginal returns to consumption) can achieve a higher utility by insuring.



- (d) Now assume that the insurance company is not able to differentiate between the three types. Therefore, it offers a policy that fully insures individuals at the same price to all three types. Assuming all three types buy this insurance policy, what is the price for insurance?

Solution: Assuming (1) firms make zero profits and (2) equal shares of each type in the population, the actuarially fair price is:

$$p = d * E[\text{Pr(disability)}] = 490 * \frac{0.05 + 0.10 + 0.60}{3} = \$122.5$$

- (e) Prove that with the price you found in part (d), the low-risk type would not be willing to buy the full insurance policy.

Solution: As you can see from part (c), the price of \$122.50 is greater than the maximum price that the low-risk type is willing to pay \$88.83. Therefore, the low-risk type will not buy the policy.

- (f) Since the low-risk type does not buy the policy, the insurance company cannot offer the policy at the price from part (c). What is the new price for this policy if the risk averse drop out of the market? Who will buy at this price?

Solution: Without the low-risk types, the actuarially fair price is now:

$$p = d * E[\text{Pr(disability)}] = 490 * \frac{0.10 + 0.60}{2} = \$171.5$$

Now the medium-risk types will refuse to buy the policy, since this is higher than their maximum (\$161.88). Therefore, only the high-risk types will buy (at their actuarially fair price of \$294) and the market has unraveled.

- (g) You have demonstrated an example of a market unraveling. Explain the intuition for why it happened.

Solution:

- This market unraveling is called adverse selection.
- It is the result of asymmetric information between the insurance companies and those it may insure.
- The insurance company cannot distinguish between risk preferences, so it must offer the same policy to everyone, but in order to earn zero profit they must charge a premium equal to the population risk times the coverage amount.
- Since the risk adverse know who they are, they will not insure in order to maximize utility.
- This causes the risk of accident/disability among the insured population to increase, causing fewer people to insure, and so on until no insurance contract is offered at all even though everybody wants full actuarially fair insurance.

- (h) How would results change if the individuals' income is zero when incapacitated?

Solution: Given that at zero income the log utility is negative infinity, people would never choose not to be insured. We therefore obtain again a pooling equilibrium and we might

not need government intervention. [Remember, a pooling equilibrium is such that insurance companies offer a contract based on average risk.]